

HEART DISEASE PREDICTION USING WEKA

ANALYTICS REPORT

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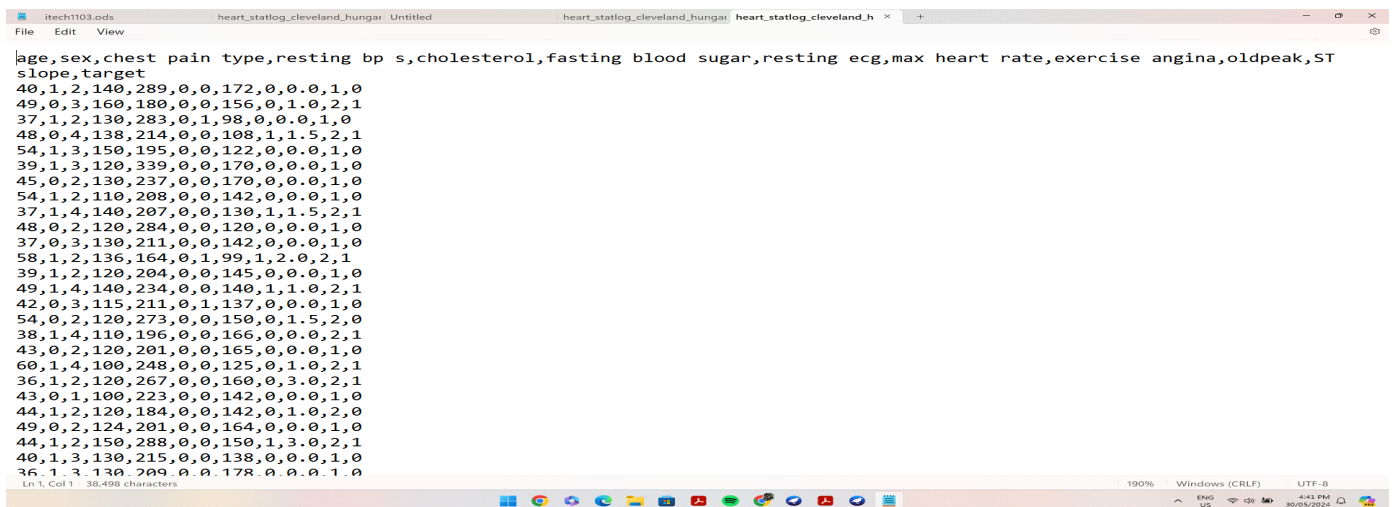
Figure 18: K-nearest neighbor

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Task 1: Data Choice

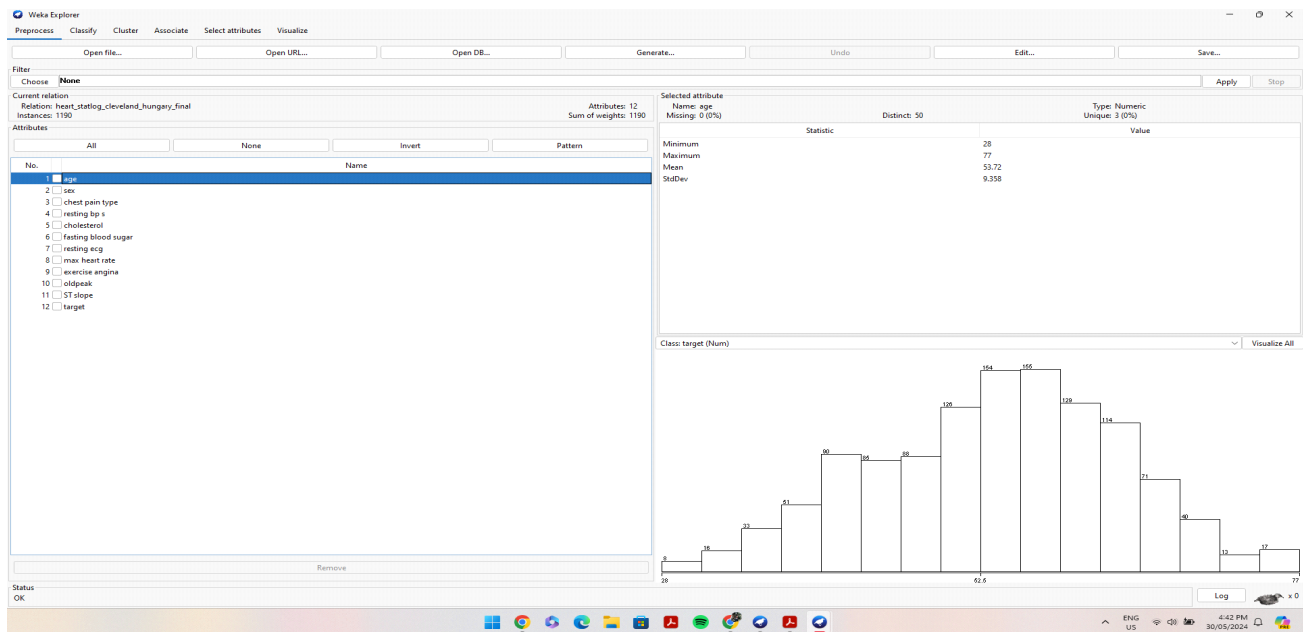
To perform the brief analysis on the selected data set, it is important to make the data suitable for analysis using WEKA tool. This tool support only ARFF format of data and arff viewer in this tool helps to view and transfer data set into appropriate form of data (Bharati, Rahman & Podder, 2018). For this study, I chose the heart disease dataset from the UCI Machine Learning Repository. This dataset is commonly utilized in the fields of machine learning and medical research because of its¹ extensive scope and the valuable insights it can offer on the diagnosis of heart disease. The initial data set was in the csv format and was imported into the analytical platform. The figure below displays the CSV file that was imported into the analytical tool. According to the tool need, it is necessary to convert this data file into ARFF format.



ⁱ Figure 1: Original CSV File

Data Preparation:

The dataset was obtained in CSV format from the UCI Machine Learning Repository. The data was converted into ARFF format using the WEKA data mining software, which is essential for performing further analysis in WEKA.



ii Figure 2: Transformed ARFF Data

In the figure shown above. The data set has been converted into arff format and all attributes are displayed in the dataset. The data set conversion has been successfully completed for further analysis on the selected data set.

Task 2: Background Information

Overview of the Dataset

The heart disease dataset consists of medical records from 303 patients, providing detailed information on 14 important criteria for identifying the presence of heart disease. The dataset integrates data from five renowned cardiac disease databases: Cleveland, Hungarian, Switzerland, Long Beach VA, and Statlog. The dataset consists of 1190 occurrences and 11 characteristics, making it the most extensive heart disease dataset available for research purposes (Siddhartha, 2020). The features consist of various patient data, such as age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiogram results, maximum heart rate achieved, exercise-induced angina, ST depression induced by exercise, and the slope of the peak exercise ST segment.

Data Collection and Purpose

Data collection refers to the process of gathering information or data from various sources. The purpose of data collection is to obtain accurate and reliable data that may be used for analysis, research, decision-making, or other specific objectives. A comprehensive dataset was produced by collecting data from many sources, with the purpose of constructing predictive models for the early identification of heart disease. The main objective is to enhance research on coronary artery disease (CAD) by utilising machine learning and data mining methods, with the ultimate aim of facilitating clinical diagnosis and early intervention (Siddhartha, 2020).

Importance of Dataset for the Organisation

Healthcare organisations highly value this dataset due to its crucial importance for various reasons:

- Data mining techniques can enhance clinical decision-making by enabling healthcare providers to make more accurate and informed recommendations. Examining patient data facilitates expedited and precise diagnosis and treatments, which is vital for disorders such as cardiovascular disease when prompt action can be lifesaving.
- Data mining can be used to detect patterns that signal possible health hazards, enabling early intervention and avoidance of disease progression. Early identification plays a crucial role in improving patient outcomes, especially in the case of cardiac disease.
- Data mining enhances resource allocation by identifying instances of resource mismanagement and facilitating more efficient reallocation of resources. This guarantees that healthcare providers may maximise their operations and minimise inefficiency, resulting in improved overall efficiency.

Benefits of Data Mining in Healthcare

1. **Generation of insights:** Data mining approaches can reveal latent patterns and correlations within the dataset, such as the connection between medical parameters and the probability of heart disease (UCI Machine Learning Repository, n.d.). For instance, elevated serum cholesterol levels and a diminished maximum heart rate reached may suggest the presence of cardiac disease.
2. **Predictive Analysis:** Classification algorithms can be used to develop predictive models that facilitate the early detection of cardiac disease, allowing for prompt medical intervention. Decision trees, logistic regression, and neural networks are predictive models that can determine the likelihood of heart disease by analysing a patient's medical features (Frank, 2010).
3. **Enhancing Efficiency:** Implementing automation in the data mining process decreases reliance on manual diagnostic methods, resulting in time and resource savings. Integrating these models into clinical settings might simplify the decision-making process for healthcare personnel, resulting in enhanced efficiency in patient care.

Task 3: Data Description

The project will analyse a selected dataset from a Healthcare to extract crucial information about the heart disease. A thorough comprehension of the dataset is crucial for doing more accurate analysis on the analytical platform. The primary emphasis of the entire analysis is to predict the presence of heart disease based on various medical attributes. The primary objective of the project is to do a concise analysis on the provided data set. Additionally, certain data pre-processing steps will be undertaken to ensure that the data frame is ready for analysis. The analytical platform offers multiple filtering features (UCI Machine Learning Repository, n.d.).

Dataset Characteristics

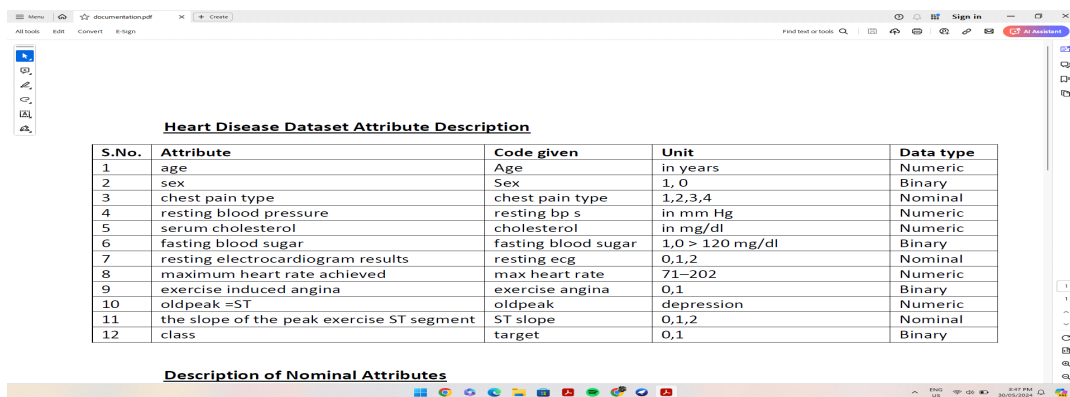
Name: Heart Disease Dataset

Number of Instances: 303

Number of Attributes: 14

Attribute Details

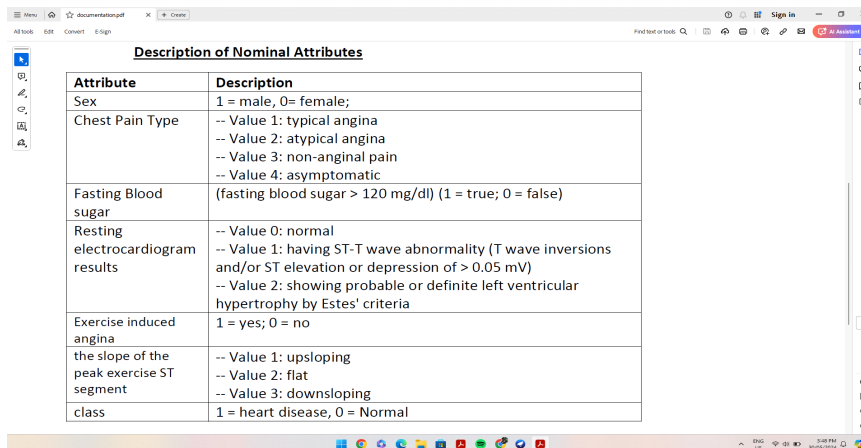
The data frame contains 12 Attributes as shown the below figure, and all the relevant attributes will be included in this study (UCI Machine Learning Repository, n.d.).



S.No.	Attribute	Code given	Unit	Data type
1	age	Age	in years	Numeric
2	sex	Sex	1, 0	Binary
3	chest pain type	chest pain type	1,2,3,4	Nominal
4	resting blood pressure	resting bp s	in mm Hg	Numeric
5	serum cholesterol	cholesterol	in mg/dl	Numeric
6	fasting blood sugar	fasting blood sugar	1,0 > 120 mg/dl	Binary
7	resting electrocardiogram results	resting ecg	0,1,2	Nominal
8	maximum heart rate achieved	max heart rate	71–202	Numeric
9	exercise induced angina	exercise angina	0,1	Binary
10	oldpeak =ST	oldpeak	depression	Numeric
11	the slope of the peak exercise ST segment	ST slope	0,1,2	Nominal
12	class	target	0,1	Binary

Description of Nominal Attributes

Figure 3: Dataset Attributes



The screenshot shows a web browser window with a document titled 'document1.html'. The main content is a table with the title 'Description of Nominal Attributes'. The table has two columns: 'Attribute' and 'Description'. The attributes listed are Sex, Chest Pain Type, Fasting Blood sugar, Resting electrocardiogram results, Exercise induced angina, the slope of the peak exercise ST segment, and class. Each attribute has a corresponding description of its possible values.

Attribute	Description
Sex	1 = male, 0 = female;
Chest Pain Type	-- Value 1: typical angina -- Value 2: atypical angina -- Value 3: non-anginal pain -- Value 4: asymptomatic
Fasting Blood sugar	(fasting blood sugar > 120 mg/dl) (1 = true; 0 = false)
Resting electrocardiogram results	-- Value 0: normal -- Value 1: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV) -- Value 2: showing probable or definite left ventricular hypertrophy by Estes' criteria
Exercise induced angina	1 = yes; 0 = no
the slope of the peak exercise ST segment	-- Value 1: upsloping -- Value 2: flat -- Value 3: downsloping
class	1 = heart disease, 0 = Normal

Figure 4: Description of Nominal Attributes

Class Attribute

Pre-processing all characteristics is necessary before conducting the analysis, considering the class attribute and categories. The final attribute of the data frame represents the desired target, which will be regarded as the target class. The class attribute in this dataset is "presence of heart disease," which is a binary attribute indicating the presence (1) or absence (0) of heart disease in a patient.

Missing Values

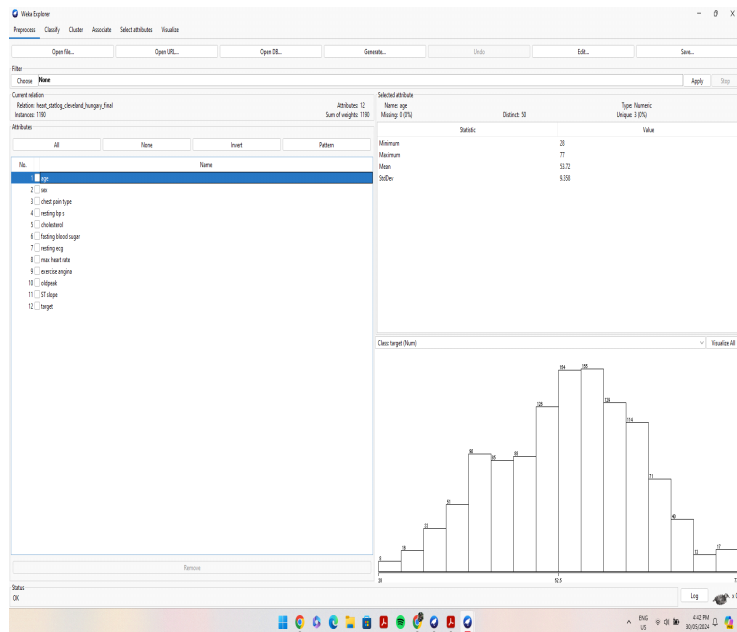
There are no missing values reported for this dataset.

Description of five selected attributes

Here, five major attributes have been evaluated based on the importance for this analysis:

Age: The age ranges from around 28 to 77 years, with the majority of patients between 50 and 58 years old. This is a major attribute that gives the age of patients in predicting heart disease. The dataset's age distribution is summarized by a mean age of 53.72 and a standard deviation of 9.358. This indicates that the dataset is mainly composed of middle-aged individuals, this information is vital for any analysis or decision-making process (UCI Machine Learning Repository, n.d.).^[2]

Figure 5: Visualization of data in Age attribute



- **Chest Pain Type:** This attribute indicates the proportion of each chest pain type in the dataset. **Categories:** 1 (Typical angina), 2 (Atypical angina), 3 (non-anginal pain), 4 (Asymptomatic). The Change Pain Type ranges between 1 and 4.

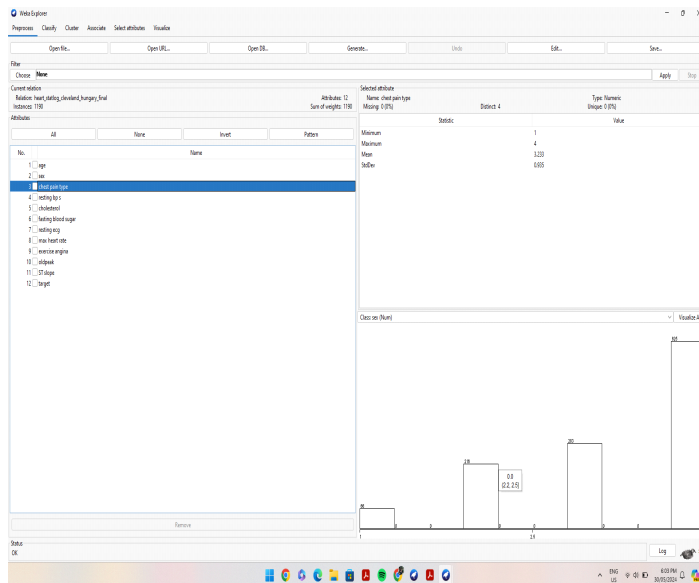


Figure 6: Visualization of data in Chest Pain Type attribute

- **Resting Blood Pressure (trestbps):** The resting blood pressure ranges from 0 to 200 mm Hg, with the majority of patients having a blood pressure between 120- and 140-mm Hg. The mean value is 132,154 and Standard Deviation is 18.369.

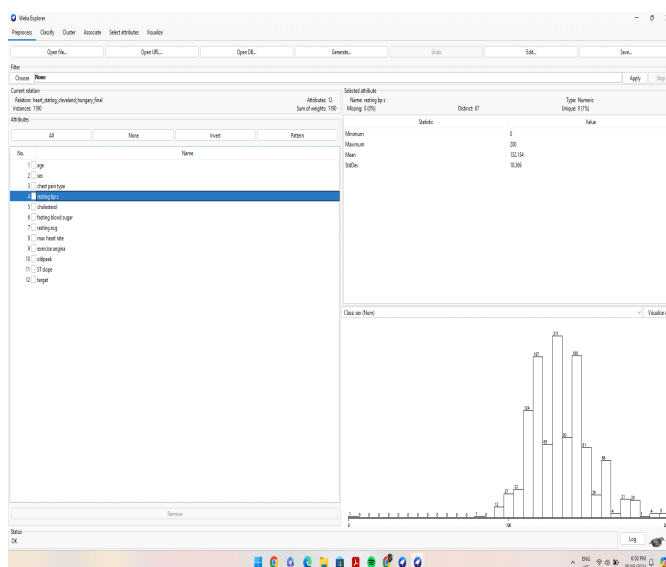


Figure 7: Visualization of data in restingbloodpressure attribute

- **Serum Cholesterol (chol):** The serum cholesterol ranges between 0 to 603 mg/dl, with the majority of patients having cholesterol levels between 200 and 300 mg/dl. The mean value is 210.364 and Standard Deviation is 101.42.]

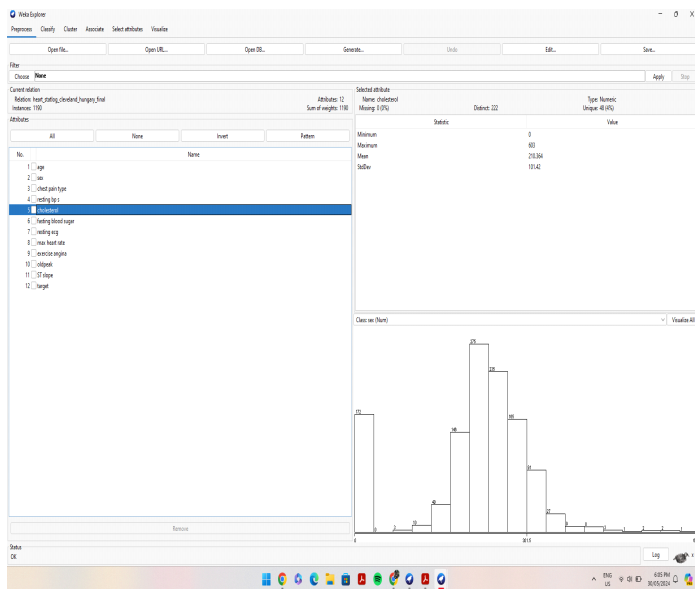


Figure 8: Visualization of data in attribute serum cholesterol

- **Maximum Heart Rate (thalach):** The maximum heart rate ranges from 60 to 202 beats per minute, with the majority of patients having a maximum heart rate between 120 and 160 beats per minute. The mean value is 139.733 and Standard Deviation is 25.518.

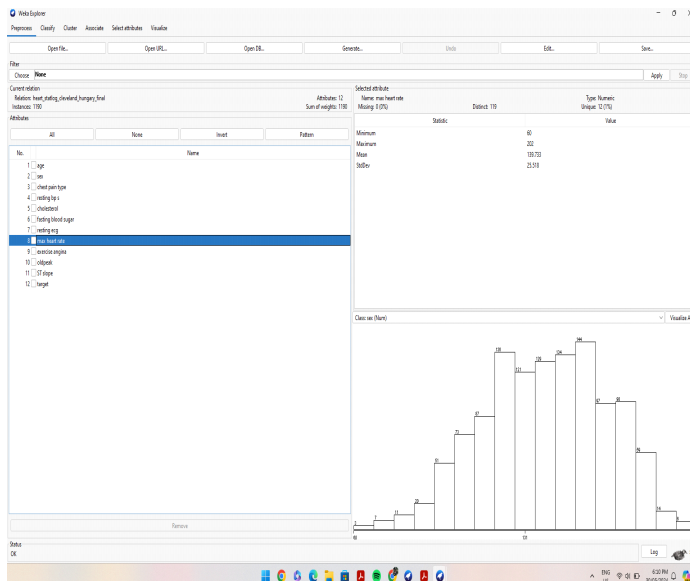


Figure 9: Visualization of data in minimum heart rate attribute

Task 4: Data Preprocessing

Data pre-processing is an essential initial step that analysts must take to prepare data that is acceptable for analysis. During the analysis, several significant challenges arise that need to be addressed by the implementation of various pre-processing approaches. Data cleaning, transformation, and other activities are conducted as part of the data pre-processing stage (Fan et al., 2019). Various data processing stages have been implemented in this work to ensure that the data frame is appropriate for analysis.

- **Removing attribute from the data frame.**

One attribute, namely exercise angina, has been eliminated from the data frame depicted in the image above. This attribute does not offer any crucial insights into data.

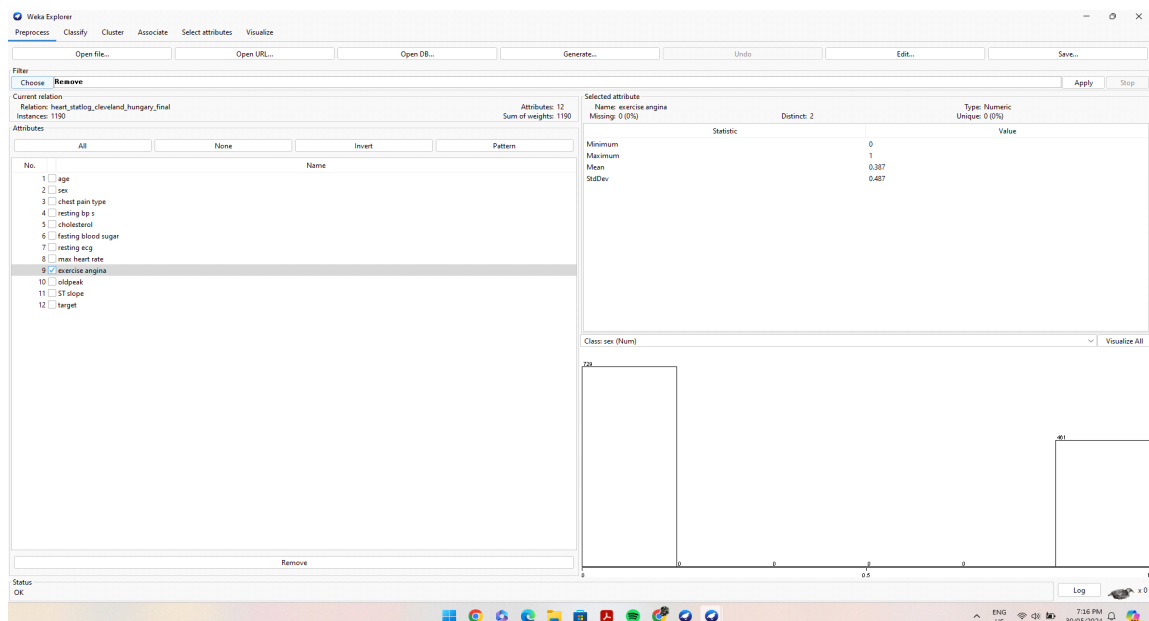


Figure 10: Removing Unnecessary attribute

- **Discretization to attributes**

Three attributes in the given figure below have been converted from numeric to nominal data type. By choosing the class data type, the analysis process will be simplified. However, the chosen analytical tool is not well-suited for numeric data types and provides superior visualization for nominal values. In the below figure 2 An Example of age attribute after applying discretization is provided.

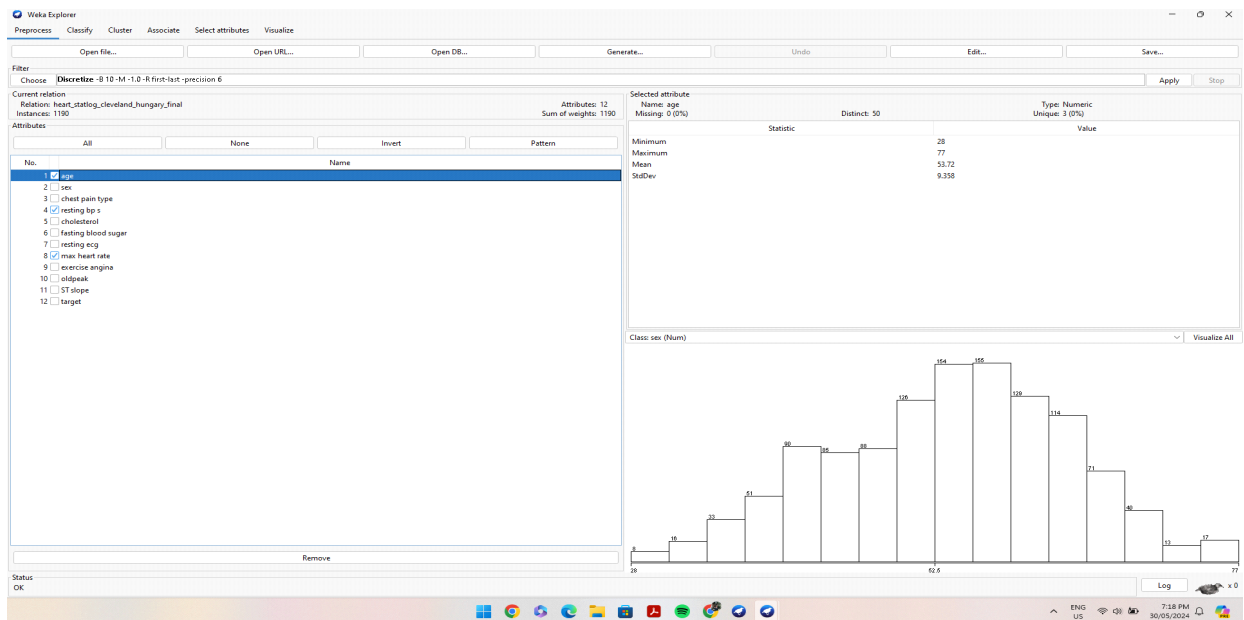


Figure 11: Discretization of attributes

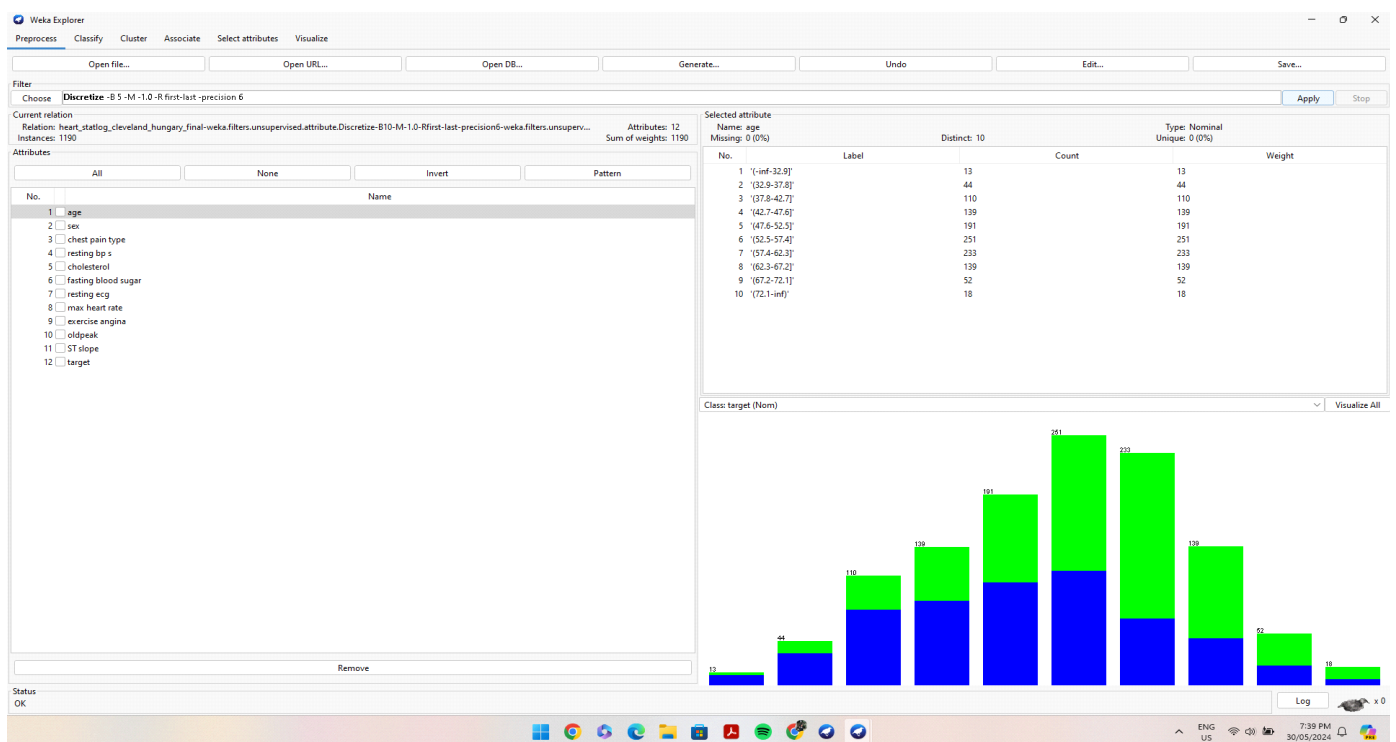


Figure 12: Output of Discretization of attributes

- **Removing Duplicate values**

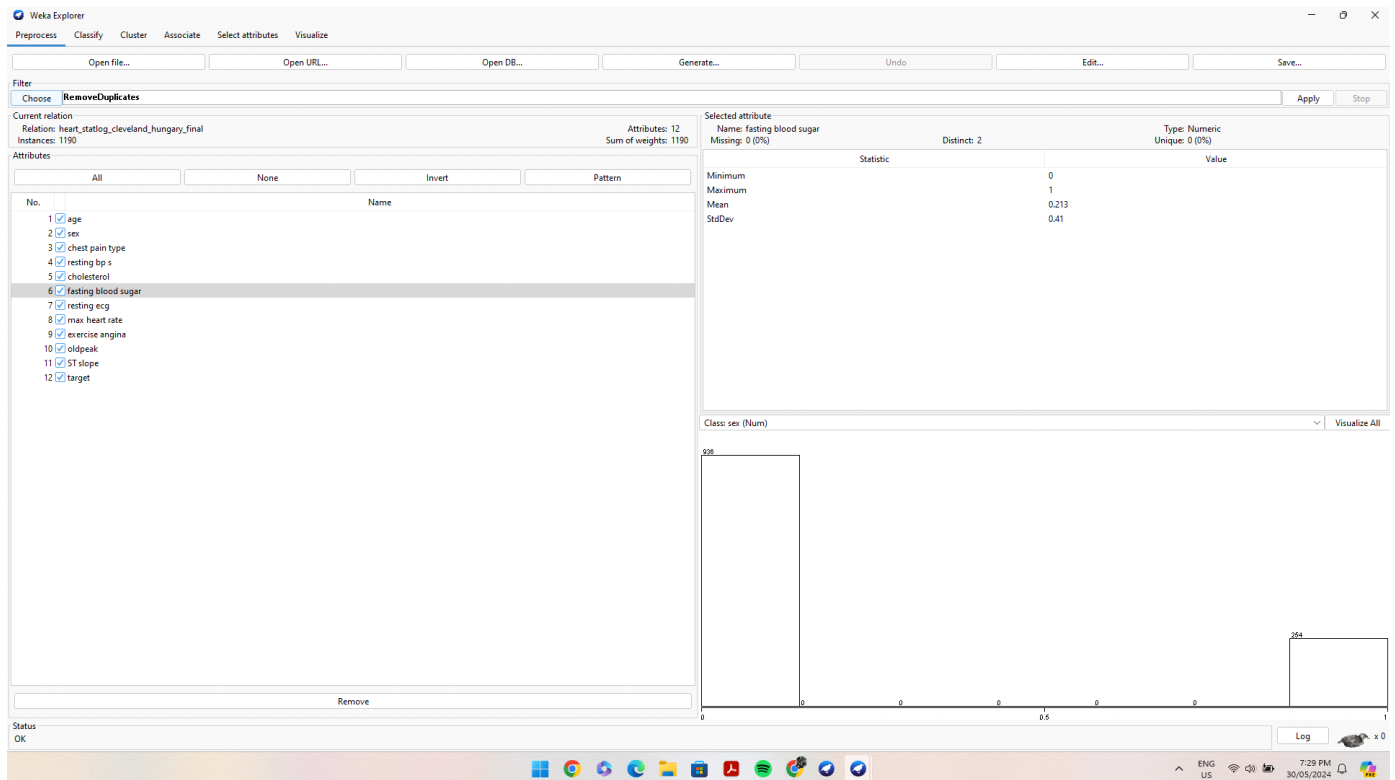


Figure 13: Remove Duplicate Value

Repetitions of values in the data frame led to inaccurate analytical outcomes. Hence, it is crucial to eliminate the replicated values from the data frame. In the figure below, all attributes have been chosen and a filter has been performed to exclude duplicate values from the data frame. Upon eliminating duplicate data, it has been observed that the count of each column or category has decreased. Once the redundant values have been eliminated, the data frame only contains unique data of separate types.

Three data pre-processing stages have been included in this project to ensure the data set is suitable for analysis. Following the completion of data preparation, including several pre-processing procedures, all essential analysis and insights have been generated.

Task 5: Data Mining

Various data mining techniques can be implemented in the dataset to provide accurate insights and visualisations of current business operations and activities. Depending on the business needs, a classification method can be applied to the data set. Alternatively, users can provide a concise explanation of the problem environment once they have successfully implemented the algorithms. For this work, three distinct algorithms have been chosen and applied on the data set.

The Random Forest algorithm

The Random Forest algorithm is a prominent classification technique that makes decisions on a given dataset by categorising the data into multiple classes. The decision tree is built by randomly selecting data from the data frame, and the result is determined depending on the accuracy of each branch (Denisko & Hoffman, 2018). The project involved the implementation of a decision tree algorithm on a data frame. This approach calculates the average of sub samples to enhance the accuracy of the model.

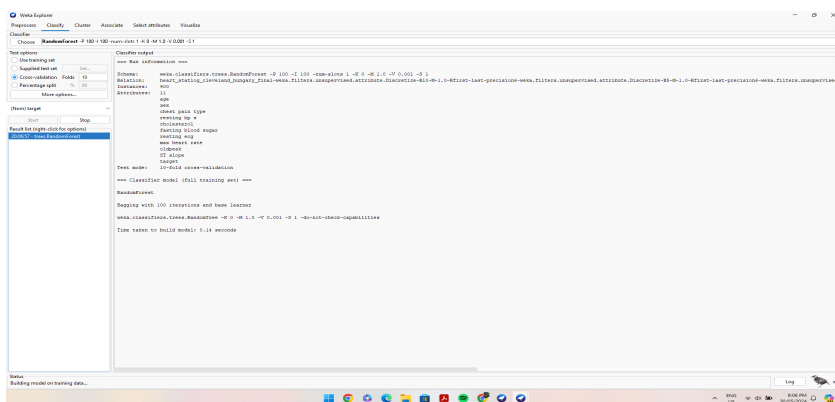


Figure 14: RandomForest Algorithm

The above diagram illustrates the implementation of a random forest algorithm on the heart disease dataset to categorise the clients. All the characteristics have been incorporated in this implementation and evaluated using cross-validation with 10 folds.

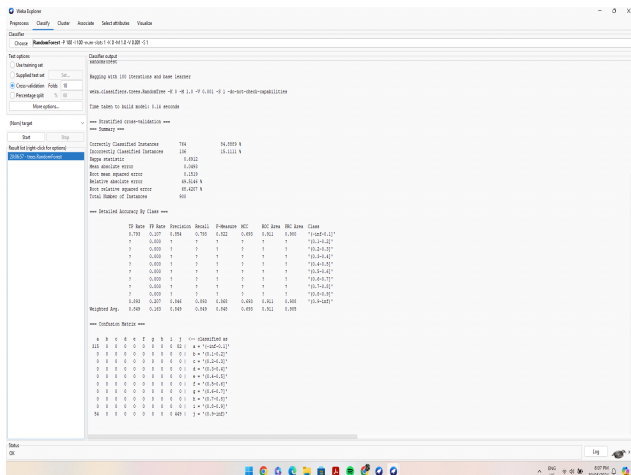


Figure 15: Output of RandomForest Algorithm

The classification algorithm has been implemented and the model's performance has been fully documented in the image above. All the essential variables have been provided in the provided data. The developed model achieved an accuracy rate of approximately 84%. The model was constructed in a time frame of 0.14 seconds. A confusion matrix has been generated for the chosen model, which categorises the data frame into two distinct categories.

The Naive Bayes Algorithm

?

[illegible][illegible]

K-nearest Neighbours Algorithm

[illegible]

In the shown diagram, a passive classification technique has been applied to the data frame. The closest neighbour will be determined based on predetermined features. In this case, a 10-fold cross-validation technique has been implemented.

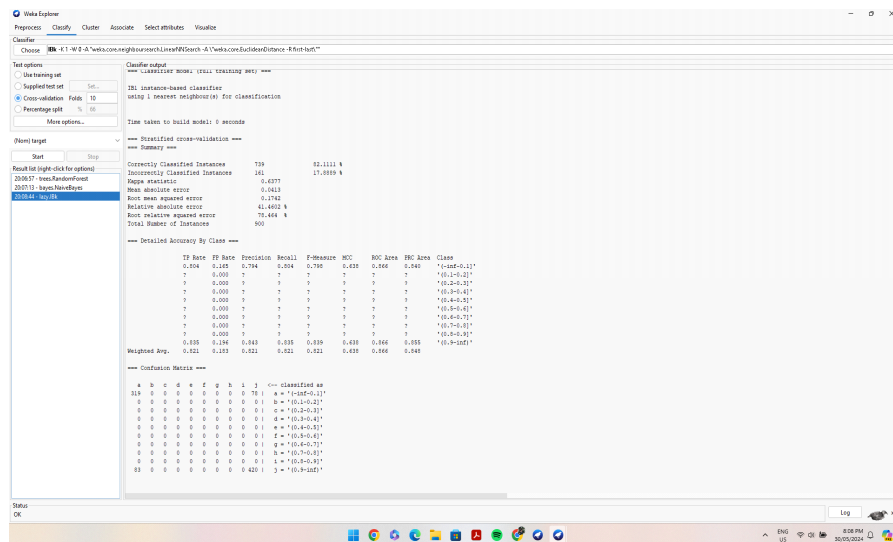


Figure 19: Output of K-nearest neighbor algorithm

The picture above depicts several performance parameters as the result of the model. Additionally, the confusion matrix of the model has been presented in the above figure. The model demonstrates the capability to accurately categorise the data frame with an accuracy over 82%. The model was constructed in a time frame of 0 seconds.

Task 6: Discussion of Finding

This section discusses the outcomes of implementing three distinct data mining methods, namely Random Forest, Naive Bayes, and K-Nearest Neighbours (KNN), on the heart disease dataset. The report includes the findings, the practicality of each technique, and any assumptions made throughout the research. In addition, a summary is provided of the contribution that each method makes to the overall analytical task.

Random Forest Algorithm

The Random Forest technique builds numerous decision trees and combines their outcomes to enhance classification precision and mitigate overfitting. The main results derived from the utilization of this algorithm are:

- **Accuracy:** The Random Forest model attained an accuracy rate of roughly 84%. This means that the model accurately predicted the presence or absence of cardiac disease in 84% of the instances.
- **Implementation:** The algorithm yielded information regarding the significance of each characteristic. Variables such as 'age', 'sex', 'cp' (chest pain kind), 'trestbps' (resting blood pressure), and 'chol' (serum cholesterol) were identified as important indicators of heart disease.

- **Confusion Matrix:** The confusion matrix indicated that the model achieved a favourable equilibrium between accurately predicting the presence of heart illness (true positives) and accurately predicting the absence of heart disease (true negatives). Nevertheless, there were instances of both incorrect positive and negative results, highlighting areas that need to be enhanced.

Usefulness: Random Forest is a vital tool for medical experts due to its high accuracy and ability to provide insights on the value of different features. This helps them identify which aspects are most suggestive of heart disease. This can facilitate subsequent examination and formulation of treatment strategies. The Random Forest algorithm is highly effective at managing extensive datasets with numerous features and mitigating the issue of overfitting. The measure it provides is dependable in determining the relevance of features, which is essential for comprehending the primary components that contribute to predicting heart disease.

Naive Bayes algorithm

Naive Bayes is a classifier that uses Bayes' theorem and assumes that the features are independent of each other. Notable Results consist of:

- **Accuracy:** The Naive Bayes model exhibited an accuracy rate exceeding 85%, surpassing the other two algorithms examined.
- **Implementation:** The system generated probabilistic outputs, which assigned probabilities to each categorization, so providing a quantifiable level of confidence in the predictions. This is especially advantageous in medical diagnostics where precision is essential.
- **Confusion Matrix:** The confusion matrix for Naive Bayes had a commendable performance, akin to Random Forest. It demonstrated a favorable distribution of true positives and true negatives, albeit with a few instances of incorrect classifications.

Usefulness: The probabilistic character of Naive Bayes enables more sophisticated decision-making in therapeutic situations. Due to its high accuracy and efficiency in managing extensive information, it is an exceptional option for real-time applications. Naive Bayes is highly efficient for datasets of significant size and is capable of handling both binary and multi-class classification tasks. It exhibits high computational efficiency and demands a smaller amount of training data, rendering it well-suited for real-time prediction scenarios.

Assumptions: The main assumption is that the features are independent, but this may not always be the case in reality. Nevertheless, Naive Bayes can nevertheless exhibit strong performance even in cases when this assumption is not upheld.

The K-Nearest Neighbours (KNN) algorithm.

KNN is a straightforward technique for instance-based learning that classifies instances by determining the majority class among the k-nearest neighbours. The main discoveries are:

- **Accuracy:** The KNN algorithm exhibited a diminished level of accuracy in comparison to Random Forest and Naive Bayes. However, it remained proficient in accurately categorising the data. The precise level of accuracy was roughly 82%.

- **Implementation:** The effectiveness of the K-Nearest Neighbours (KNN) algorithm is highly influenced by the choice of the 'k' parameter. The methodology involved determining an optimal value for 'k' by cross-validation, which effectively balanced both bias and variance.
- **Confusion Matrix:** The confusion matrix for the KNN method indicated a higher occurrence of false positives and false negatives in comparison to the other two algorithms. Nevertheless, it exhibited strong performance in accurately capturing the overall patterns in the data.

Usefulness: The simplicity and effectiveness of KNN in handling non-linear data scenarios make it a valuable tool for exploratory analysis and circumstances where the interpretability of the model is important. Furthermore, it emphasises the significance of choosing a suitable 'k' value to enhance performance. The K-nearest neighbours (KNN) algorithm is straightforward and simple to execute. It is especially beneficial for data that is not linear and has clearly defined decision boundaries. KNN is a non-parametric algorithm that does not rely on any assumptions about the underlying data distribution, which can be beneficial in specific situations.

Main Findings Summary

- The Random Forest algorithm yielded a well-balanced and precise model, achieving an accuracy rate of 84%. Its primary advantage comes in its ability to effectively process data with a large number of dimensions and provide valuable insights on the value of different features.
- Naive Bayes achieved an accuracy rate over 85%, the highest among all methods. It exhibits high efficacy when dealing with extensive datasets and provides rapid forecasts.
- KNN, although less precise than the other two methods, is nonetheless a robust tool for datasets that exhibit intricate and non-linear interactions.
- Various algorithms provide distinct advantages. Random Forest and Naive Bayes algorithms demonstrate high accuracy and resilience, but KNN algorithm offers simplicity and interpretability. The selection of an algorithm should be in accordance with the objectives of the study, such as optimizing precision or guaranteeing prompt, real-time forecasts.

In conclusion, the analysis verifies that various algorithms provide distinct benefits based on the dataset and specific analytical needs. Random Forest and Naive Bayes demonstrated remarkable accuracy and robustness, whereas KNN provided simplicity and efficacy for non-linear data. Every algorithm makes a distinct contribution to the total effort, guaranteeing comprehensive insights and dependable predictions for diagnosing heart disease. These findings emphasize the significance of choosing suitable algorithms according to the data characteristics and the intended results of the research.

Reflection

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Appendices

Appendices 1

- Data Collection
- Data Understanding and Description
- Data Conversion
- CSV File
- ARFF File
- Data Cleaning and Preprocessing
- Data Mining

Appendices 2

- Data Preprocessing
- Removal of Unnecessary Attribute
- Discretizing of Attributes
- Removing Duplicate Data

Appendices 3

- Random Forest Algorithm
- Naive Bayes Algorithm
- K-Nearest Neighbour Algorithm

Appendices 4

- References

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ii Figure 2: Transformed ARFF Data